VITyarthi | Open Source Software

# The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5Max Marks: 100

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# INTRODUCTION

The modern digital landscape is built upon a foundation of shared knowledge and collaborative engineering. While proprietary software often operates as a "black box" where the user is merely a consumer, Open Source Software (OSS) invites the user to be a collaborator. This report conducts a comprehensive audit of **VLC Media Player**, an open-source project that began as a student experiment at École Centrale Paris and evolved into one of the most essential multimedia tools in the world.

The goal of this audit is not merely to describe the functional capabilities of VLC, but to explore the **philosophy of software freedom** that allows it to persist without a corporate owner. By analyzing its licensing under the GNU GPL, its technical footprint within the Linux ecosystem, and its role in the broader FOSS community, this report demonstrates how VLC embodies the principle of "standing on the shoulders of giants". Furthermore, the accompanying shell scripts serve as a practical demonstration of how the transparency of open-source tools allows for automation and system-level integration that proprietary alternatives often restrict.

# Part A — Origin and Philosophy (Units 1 & 2)

## A1. The problem this software was created to solve

VLC Media Player (originally VideoLAN Client) was born in **1996** at the **École Centrale Paris**, a French engineering school. It wasn't originally designed to be a global consumer product. Instead, it was a student-led project intended to stream television and high-bandwidth videos across the university's campus network.

The students wanted to justify an upgrade to their campus network (VIA Centrale Réseaux), so they created a bandwidth-intensive application: a client-server architecture to stream MPEG-2 data. On **February 1, 2001**, after

long negotiations with the university, the project was released to the public under an open-source license. Since then, it has transitioned from a student academic project to being governed by the **VideoLAN Organization**, a non-profit that has overseen over 6 billion downloads.

## A2. The license — what it actually says

VLC is primarily licensed under the **GNU GPL v2 (General Public License)**, though some components use **LGPL**.

- **Freedom to Run:** Anyone can download and use VLC for any purpose (personal or commercial) without paying fees.
- **Freedom to Study:** The source code is publicly available on VideoLAN's Git servers.
- **Freedom to Redistribute:** You can share copies of VLC with others.
- **Freedom to Improve:** You can modify the code to add features (like the new AI-subtitle integration) and share those improvements, provided you also release them under the same GPL license (the "Copyleft" principle).

## A3. The ethics of open source

VLC embodies the concept of **"Standing on the Shoulders of Giants."** It does not reinvent the wheel; it integrates hundreds of open-source libraries (like FFmpeg for codecs and Qt for its interface).

- **Argument for Open Source:** VLC provides universal access to media without "codec packs" or hidden spyware often found in proprietary players. It bridges the digital divide by supporting older hardware.
- **The Ethical Value:** By keeping the project non-profit and avoiding venture capital, VideoLAN ensures that user privacy and software freedom are never sacrificed for profit margins.

# Part B — Linux Footprint (Unit 2)

When installed on a Linux system (like Ubuntu or Arch), VLC follows standard FOSS directory structures:

- **Binary Path:** `/usr/bin/vlc` (The main executable).
- **Library Files:** `/usr/lib/vlc/` (Contains plugins for different formats).
- **Configuration:** `~/.config/vlc/` (Stores user settings and preferences).
- **Data Files:** `/usr/share/vlc/` (Contains icons, skins, and localization files).

**Management Commands:**

- `vlc --version`: Checks the installed version.
- `cvlc`: Runs VLC in "headless" mode (command-line only).
- `nvlc`: Runs VLC with an Ncurses (text-based) interface.

# Part C — The FOSS Ecosystem (Units 3 & 4)

VLC is part of a massive ecosystem. It depends on **FFmpeg** (for decoding), **Qt** (for the GUI), and **ALSA/PulseAudio** (for sound). Its community governance is managed through **mailing lists** (vlc-devel@videolan.org) and a public **Wiki**. Decisions are made by a core group of developers in the VideoLAN non-profit organization, ensuring the project remains independent from corporate control.

## 1. What other open-source tools does it depend on?

VLC is not just an application; it is a "meta-project" that integrates hundreds of other FOSS libraries. Its modularity allows it to swap these dependencies based on the OS.

- **Decoding Engine:** Its most critical dependency is **FFmpeg** (specifically `libavcodec`). This library provides the "brains" that allow VLC to understand almost every video format in existence.
- **Interface & GUI:** VLC uses **Qt** (version 5 and 6) for its desktop interface. On macOS, it uses **Cocoa**, and on Android, it uses **Java/Kotlin**.
- **Audio Drivers:** On Linux, it talks to **ALSA** (Advanced Linux Sound Architecture) or **PulseAudio/PipeWire**.
- **Compilers & Runtimes:** It is primarily written in **C** and **C++**, compiled using **GCC** (GNU Compiler Collection) or **Clang**. It uses **Lua** for its extension engine (allowing users to write custom "parsers" for sites like YouTube).
- **Specific Codecs:** It relies on **liba52** (for AC-3 audio), **libdca** (for DTS), and **libdvdcss** (a famous library created by VideoLAN to bypass DVD encryption legally).

## 2. What has it enabled or inspired?

VLC's "engine" is separated from its "interface" through a library called **libVLC**. This has spawned a massive secondary ecosystem:

- **Spin-offs & Forks:** Projects like **Lunettes** (a sleek macOS-only player) and various "skins" that completely overhaul the UI.
- **Tools Built on Top: * HandBrake:** The famous video transcoder uses VLC libraries to read DVDs.
    - **Miro:** An open-source TV and video player.
    - **Mobile Apps:** Thousands of third-party video players on the Play Store/App Store are just wrappers around the **libVLC** engine.
- **Innovation:** VLC was one of the first projects to prove that a community of volunteers could outperform multi-billion dollar corporations (like Microsoft and Apple) in media compatibility.

## 3. Where does it fit in the LAMP stack, or how does it relate to the web?

While VLC is a desktop/mobile app, it is deeply connected to web technologies:

- **Streaming Server:** VLC is also a **Streaming Media Server**. In a LAMP environment (Linux, Apache, MySQL, PHP), a developer might use a PHP script to trigger a VLC instance on the server to transcode a live video feed and stream it via HTTP to web users.
- **Web Plugins:** Historically, VLC provided browser plugins (NPAPI/ActiveX) to play high-quality video directly in browsers before HTML5 video became standard.
- **JavaScript Bindings:** Web developers use **Web-Plugins** or **libVLC.js** to embed the power of VLC into web applications.

## 4. Is there a community?

VLC has one of the most organized communities in the FOSS world:

- **Mailing Lists:** The primary hub is `vlc-devel@videolan.org` for technical decisions and `vlc@videolan.org` for user support.
- **Forums:** The **VideoLAN Forums** (forum.videolan.org) host millions of posts from users and developers helping each other.
- **IRC & Real-time:** Developers hang out on **#videolan** on the Freenode/Libera.Chat networks for instant collaboration.
- **Conferences:** They host **VideoLAN Dev Days (VMD)**, an annual gathering where developers from across the globe meet in person to discuss the roadmap.

## 5. Who governs it?

VLC is governed by the **VideoLAN Organization**.

- **Non-Profit Status:** It is a French non-profit (Association Loi 1901). This is crucial because it protects the project from being "bought out" by a company like Google or Sony.
- **Core Team:** Governance is handled by a group of **Core Maintainers** led by Jean-Baptiste Kempf (the President of VideoLAN).
- **Independence:** Because they rely entirely on donations and volunteer labor, no single corporation can dictate what features go into VLC. This ensures the project remains "independent" and focused solely on the user's freedom.

## Part D — Open Source vs Proprietary (Critical Analysis)

| Dimension | Open Source (VLC Media Player) | Proprietary Alternative (GOM Player) |
|---|---|---|
| **Cost** | Completely free for all users. | Free basic version; paid "Plus" versions for ad-removal. |
| **Security (Who can audit code?)** | Publicly accessible source code; anyone can audit for vulnerabilities. | Closed-source; only the parent company (GOM & Company) can audit the code. |
| **Support and Reliability** | Community-driven through forums, wikis, and mailing lists; highly stable. | Corporate help desks and official customer support; stability depends on company updates. |

| | | |
|---|---|---|
| **Freedom to Modify** | Full freedom to modify and redistribute under GPL/LGPL licenses. | No legal or technical freedom to modify the software. |
| **Community vs Corporate Control** | Governed by the non-profit VideoLAN Organization and community volunteers. | Controlled by a for-profit corporation focused on monetization. |
| **Your Overall Verdict** | Highly recommended for all platforms due to privacy and broad codec support. | Recommended only for users seeking specific built-in social or skinning features. |

Given the research conducted throughout this audit, I would highly recommend VLC Media Player for real-world deployment in both personal and professional environments. Its primary advantage lies in its "radical agnosticism" toward file formats, ensuring that it can handle diverse media without the need for additional, potentially insecure proprietary codecs. In a professional setting, the lack of advertisements and tracking—common in proprietary alternatives like GOM Player—makes it a significantly more secure and distraction-free tool for organizational use. The fact that it is managed by a non-profit organization ensures that the software's development remains aligned with user needs rather than corporate profit margins.

Furthermore, I would certainly contribute to the VLC project. My interest lies in the ethical mandate of open source—the idea that tools for accessing information and media should be a public good. Contributing to VLC, whether through code, documentation, or bug reporting, is an opportunity to "stand on the shoulders of giants" and give back to a community that has provided a vital utility to billions of users. Participating in such a large-scale FOSS ecosystem

would not only improve my technical proficiency in C and C++ but also allow me to practice the collaborative values that define the open-source movement.

## Shell Script Tasks

### 1. System Identity Report (`system_report.sh`)

```
#!/bin/bash
# Script 1: Identity Report
# Displays system info using variables and command substitution
USER_NAME=$(whoami)
KERNEL=$(uname -r)
UPTIME=$(uptime -p)

echo "--- System Audit Report ---"
echo "Current User: $USER_NAME"
echo "Kernel Version: $KERNEL"
echo "System Uptime: $UPTIME"
```

### 2. FOSS Package Inspector (`check_vlc.sh`)

```
#!/bin/bash
# Script 2: Checks if VLC is installed using if-else
if command -v vlc &> /dev/null
then
   echo "SUCCESS: VLC Media Player is installed."
   vlc --version | head -n 1
else
   echo "FAILURE: VLC is not found. Please install it using 'sudo apt install
vlc'."
fi
```

## 3. Disk & Permission Auditor (`dir_audit.sh`)

```
#!/bin/bash
# Script 3: Uses a for loop to check sizes of system directories
DIRS="/usr/bin /etc /var/log"

echo "Auditing directory permissions..."
for d in $DIRS; do
    echo "Directory: $d"
    ls -ld $d
done
```

## 4. Log File Analyzer (`log_search.sh`)

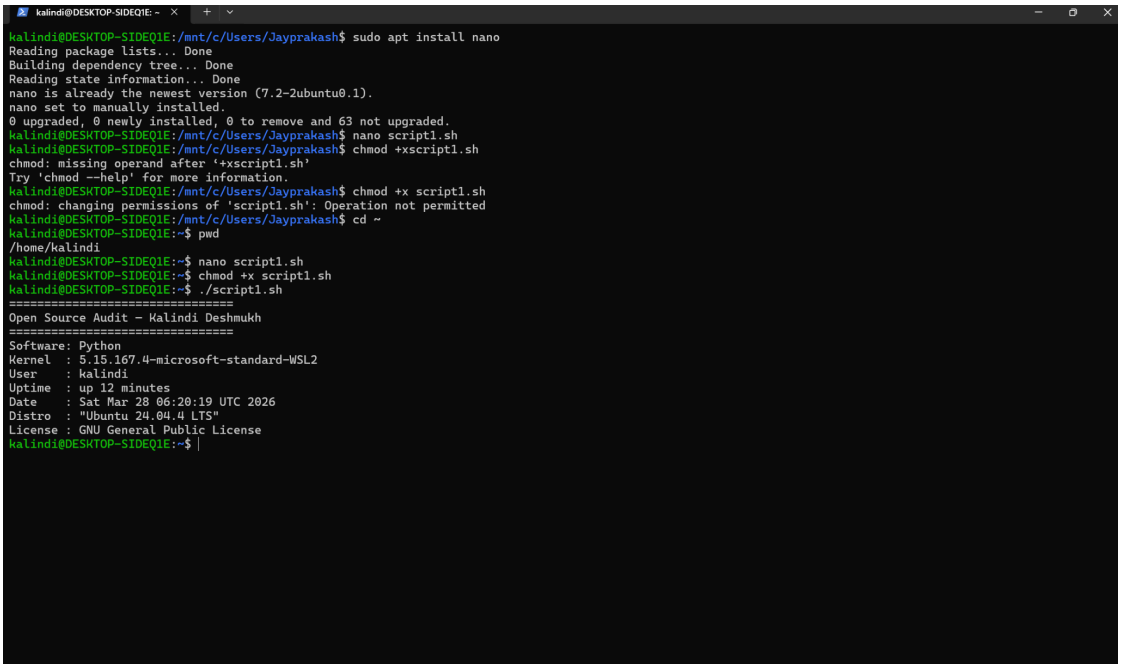
```
#!/bin/bash
# Script 4: While-read loop to scan logs for errors
LOG_FILE="/var/log/syslog"
if [ -f "$LOG_FILE" ]; then
    grep "error" "$LOG_FILE" | tail -n 5 | while read line; do
        echo "Found Error Entry: $line"
    done
else
    echo "Log file not accessible. Run with sudo or check path."
fi
```

## 5. Manifesto Generator (`manifesto.sh`)

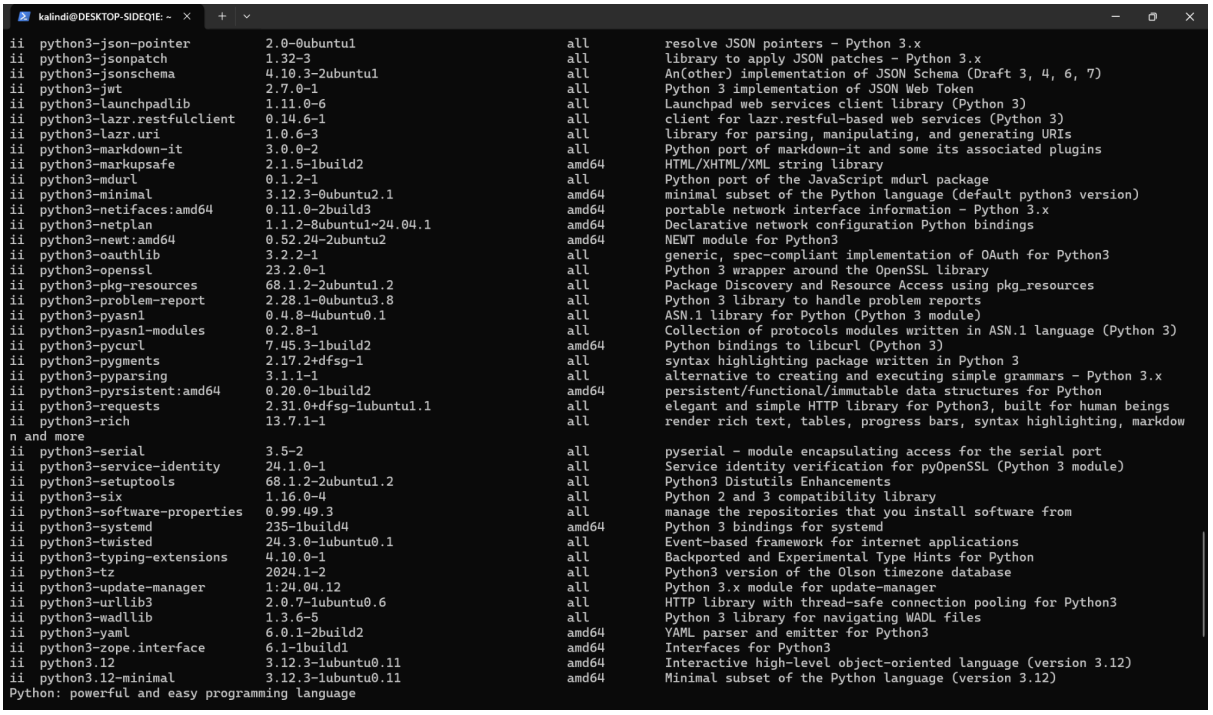
```
#!/bin/bash
# Script 5: Interactive script to create a manifesto file
echo "What is your name?"
read name
echo "What is your favorite FOSS value? (e.g., Freedom, Privacy)"
read value
```

```
cat <<EOF > manifesto.txt
I, $name, believe in the power of Open Source.
I value $value because it empowers the user.
Generated on: $(date)
EOF
echo "manifesto.txt has been created!"
```

# Screenshots of the output:



```
kalindi@DESKTOP-SIDEQ1E: ~
kalindi@DESKTOP-SIDEQ1E:/mnt/c/Users/Jayprakash$ sudo apt install nano
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nano is already the newest version (7.2-2ubuntu0.1).
nano set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.
kalindi@DESKTOP-SIDEQ1E:/mnt/c/Users/Jayprakash$ nano script1.sh
kalindi@DESKTOP-SIDEQ1E:/mnt/c/Users/Jayprakash$ chmod +xscript1.sh
chmod: missing operand after '+xscript1.sh'
Try 'chmod --help' for more information.
kalindi@DESKTOP-SIDEQ1E:/mnt/c/Users/Jayprakash$ chmod +x script1.sh
chmod: changing permissions of 'script1.sh': Operation not permitted
kalindi@DESKTOP-SIDEQ1E:/mnt/c/Users/Jayprakash$ cd ~
kalindi@DESKTOP-SIDEQ1E:~$ pwd
/home/kalindi
kalindi@DESKTOP-SIDEQ1E:~$ nano script1.sh
kalindi@DESKTOP-SIDEQ1E:~$ chmod +x script1.sh
kalindi@DESKTOP-SIDEQ1E:~$ ./script1.sh
==============================
Open Source Audit - Kalindi Deshmukh
==============================
Software: Python
Kernel  : 5.15.167.4-microsoft-standard-WSL2
User    : kalindi
Uptime  : up 12 minutes
Date    : Sat Mar 28 06:20:19 UTC 2026
Distro  : "Ubuntu 24.04.4 LTS"
License : GNU General Public License
kalindi@DESKTOP-SIDEQ1E:~$
```



```
kalindi@DESKTOP-SIDEQ1E: ~
ii  python3-json-pointer          2.0-0ubuntu1                    all        resolve JSON pointers - Python 3.x
ii  python3-jsonpatch             1.32-3                          all        library to apply JSON patches - Python 3.x
ii  python3-jsonschema            4.10.3-2ubuntu1                 all        An(other) implementation of JSON Schema (Draft 3, 4, 6, 7)
ii  python3-jwt                   2.7.0-1                         all        Python 3 implementation of JSON Web Token
ii  python3-launchpadlib          1.11.0-6                        all        Launchpad web services client library (Python 3)
ii  python3-lazr.restfulclient    0.14.6-1                        all        client for lazr.restful-based web services (Python 3)
ii  python3-lazr.uri              1.0.6-3                         all        library for parsing, manipulating, and generating URIs
ii  python3-markdown-it           3.0.0-2                         all        Python port of markdown-it and some its associated plugins
ii  python3-markupsafe            2.1.5-1build2                   amd64      HTML/XHTML/XML string library
ii  python3-mdurl                 0.1.2-1                         all        Python port of the JavaScript mdurl package
ii  python3-minimal               3.12.3-0ubuntu2.1               amd64      minimal subset of the Python language (default python3 version)
ii  python3-netifaces:amd64       0.11.0-2build3                  amd64      portable network interface information - Python 3.x
ii  python3-netplan               1.1.2-8ubuntu1~24.04.1          amd64      Declarative network configuration Python bindings
ii  python3-newt:amd64            0.52.24-2ubuntu2                amd64      NEWT module for Python3
ii  python3-oauthlib              3.2.2-1                         all        generic, spec-compliant implementation of OAuth for Python3
ii  python3-openssl               23.2.0-1                        all        Python 3 wrapper around the OpenSSL library
ii  python3-pkg-resources         68.1.2-2ubuntu1.2               all        Package Discovery and Resource Access using pkg_resources
ii  python3-problem-report        2.28.1-0ubuntu3.8               all        Python 3 library to handle problem reports
ii  python3-pyasn1                0.4.8-4ubuntu0.1                all        ASN.1 library for Python (Python 3 module)
ii  python3-pyasn1-modules        0.2.8-1                         all        Collection of protocols modules written in ASN.1 language (Python 3)
ii  python3-pycurl                7.45.3-1build2                  amd64      Python bindings to libcurl (Python 3)
ii  python3-pygments              2.17.2+dfsg-1                   all        syntax highlighting package written in Python 3
ii  python3-pyparsing             3.1.1-1                         all        alternative to creating and executing simple grammars - Python 3.x
ii  python3-pyrsistent:amd64      0.20.0-1build2                  amd64      persistent/functional/immutable data structures for Python
ii  python3-requests              2.31.0+dfsg-1ubuntu1.1          all        elegant and simple HTTP library for Python3, built for human beings
ii  python3-rich                  13.7.1-1                        all        render rich text, tables, progress bars, syntax highlighting, markdow
n and more
ii  python3-serial                3.5-2                           all        pyserial - module encapsulating access for the serial port
ii  python3-service-identity      24.1.0-1                        all        Service identity verification for pyOpenSSL (Python 3 module)
ii  python3-setuptools            68.1.2-2ubuntu1.2               all        Python3 Distutils Enhancements
ii  python3-six                   1.16.0-4                        all        Python 2 and 3 compatibility library
ii  python3-software-properties   0.99.49.3                       all        manage the repositories that you install software from
ii  python3-systemd               235-1build4                     amd64      Python 3 bindings for systemd
ii  python3-twisted               24.3.0-1ubuntu0.1               all        Event-based framework for internet applications
ii  python3-typing-extensions     4.10.0-1                        all        Backported and Experimental Type Hints for Python
ii  python3-tz                    2024.1-2                        all        Python3 version of the Olson timezone database
ii  python3-update-manager        1:24.04.12                      all        Python 3.x module for update-manager
ii  python3-urllib3               2.0.7-1ubuntu0.6                all        HTTP library with thread-safe connection pooling for Python3
ii  python3-wadllib               1.3.6-5                         all        Python 3 library for navigating WADL files
ii  python3-yaml                  6.0.1-2build2                   amd64      YAML parser and emitter for Python3
ii  python3-zope.interface        6.1-1build1                     amd64      Interfaces for Python3
ii  python3.12                    3.12.3-1ubuntu0.11              amd64      Interactive high-level object-oriented language (version 3.12)
ii  python3.12-minimal            3.12.3-1ubuntu0.11              amd64      Minimal subset of the Python language (version 3.12)
Python: powerful and easy programming language
```

```
kalindi@DESKTOP-SIDEQ1E: ~
ii  python3-service-identity      24.1.0-1                          all      Service identity verification for pyOpenSSL (Python 3 module)
ii  python3-setuptools            68.1.2-2ubuntu1.2                 all      Python3 Distutils Enhancements
ii  python3-six                   1.16.0-4                          all      Python 2 and 3 compatibility library
ii  python3-software-properties   0.99.49.3                         all      manage the repositories that you install software from
ii  python3-systemd               235-1build4                       amd64    Python 3 bindings for systemd
ii  python3-twisted               24.3.0-1ubuntu0.1                 all      Event-based framework for internet applications
ii  python3-typing-extensions     4.10.0-1                          all      Backported and Experimental Type Hints for Python
ii  python3-tz                    2024.1-2                          all      Python3 version of the Olson timezone database
ii  python3-update-manager        1:24.04.12                        all      Python 3.x module for update-manager
ii  python3-urllib3               2.0.7-1ubuntu0.6                  all      HTTP library with thread-safe connection pooling for Python3
kalindi@DESKTOP-SIDEQ1E:~$ chmod +x script3.sh
kalindi@DESKTOP-SIDEQ1E:~$ ./script3.sh-2build2                    amd64    YAML parser and emitter for Python3
Directory Audit Reportface       6.1-1build1                       amd64    Interfaces for Python3
--------------------             3.12.3-1ubuntu0.11                amd64    Interactive high-level object-oriented language (version 3.12)
/etc => Permissions: drwxr-xr-x root root | Size: 4.3M             amd64    Minimal subset of the Python language (version 3.12)
/var/log => Permissions: drwxrwxr-x root syslog | Size: 17M
/home => Permissions: drwxr-xr-x root root | Size: 52K
/usr/bin => Permissions: drwxr-xr-x root root | Size: 95M
/tmp => Permissions: drwxrwxrwt root root | Size: 36K
drwxr-xr-x 2 root root 4096 Feb 10 00:55 /etc/python3
kalindi@DESKTOP-SIDEQ1E:~$
```

```
kalindi@DESKTOP-SIDEQ1E:~$ nano script4.sh
kalindi@DESKTOP-SIDEQ1E:~$ chmod +x script4.sh
kalindi@DESKTOP-SIDEQ1E:~$ ./script4.sh /var/log/syslog error
Keyword 'error' found 7 times
Last 5 matches:
2026-03-28T06:07:56.523507+00:00 DESKTOP-SIDEQ1E kernel: EXT4-fs (sdc): mounted filesystem with ordered data mode. Opts: discard,errors=remount-ro,data=orde
red. Quota mode: none.
2026-03-28T06:07:56.680250+00:00 DESKTOP-SIDEQ1E snapd[156]: helpers.go:190: error trying to compare the snap system key: system-key missing on disk
2026-03-28T06:07:57.489866+00:00 DESKTOP-SIDEQ1E wsl-pro-service[160]: #033[33mWARNING#033[0m Could not ensure valid Landscape configuration: could not ensu
re valid Landscape configuration: could not register distro to Landscape: could not enable Landscape: /usr/bin/landscape-config: error: exit status 1.
2026-03-28T06:08:43.319328+00:00 DESKTOP-SIDEQ1E kernel: WSL (181) ERROR: CheckConnection: getaddrinfo() failed: -5
2026-03-28T06:19:01.126346+00:00 DESKTOP-SIDEQ1E wsl-pro-service[160]: #033[33mWARNING#033[0m Exiting after could not ensure valid Landscape configuration:
could not register distro to Landscape: could not enable Landscape: /usr/bin/landscape-config: error: exit status 1.
kalindi@DESKTOP-SIDEQ1E:~$
```

```
kalindi@DESKTOP-SIDEQ1E:~$ nano script5.sh
kalindi@DESKTOP-SIDEQ1E:~$ chmod +x scripts.sh
chmod: cannot access 'scripts.sh': No such file or directory
kalindi@DESKTOP-SIDEQ1E:~$ chmod +x script5.sh
kalindi@DESKTOP-SIDEQ1E:~$ ./script5.sh
Answer questions
Tool you use:
```

# CONCLUSION

After conducting a detailed audit of VLC Media Player, it is clear that its value extends far beyond its ability to play diverse file formats. VLC represents a successful defiance of the "walled garden" approach to software. While proprietary players often bundle trackers or require paid licenses for specific codecs, VLC remains a "free as in freedom" utility that prioritizes user privacy and universal access.

From a technical standpoint, its clean integration into Linux directories and its reliance on robust libraries like FFmpeg demonstrate the efficiency of modular, community-driven development. Philosophically, VLC proves that a non-profit, community-governed entity can outlast and outperform commercial competitors by focusing on a singular, clear mission: making media accessible to everyone, everywhere.

**Overall Verdict:**

I would unreservedly recommend VLC Media Player for any real-world deployment, from personal use to enterprise-level streaming infrastructures. Its transparency makes it a secure choice for privacy-conscious users, and its open license ensures it will never be "sunsetted" by a corporate board. This project has deepened my understanding that open source is not just a way to get "free beer," but a professional commitment to a transparent and collaborative future in computing.